Teacher answer key: Primary Six Mathematics | Term 1 | Week 1 | Lesson 1: Types of Sets

Answers to practice

- 1. {2, 4, 6, 8}.
- 2. Yes; both contain the same three elements.
- 3. Yes; both have three elements.
- 4. No; their cardinalities are 2 and 3.
- 5. For example, {blue, red}.
- 6. 3.
- 7. For example, {1, 2} and {a, b}.
- 8. Not equal; equivalent (three elements each).

Check of worked examples

- 1. Yes. Both contain 4, 6 and 8; their order is different but their elements are the same.
- 2. Yes. Each set has three elements.
- 3. n(K) = 3. Repeated elements are counted once.

Teacher answer key: Primary Six Mathematics | Term 1 | Week 1 | Lesson 2: Intersecting, Disjoint, Union and Universal Sets

Answers to practice

- 1. {4, 8}.
- 2. {2, 4, 6, 8, 10}.
- 3. Yes.
- 4. Yes; they share 3.
- 5. Yes.
- 6. {green}.
- 7. {2, 4}.
- 8. Two overlapping circles with 5 in the overlap.

Check of worked examples

- 1. The only common member is b. Answer: {b}.
- 2. {1, 3, 5, 7}; write 3 only once.
- 3. Yes. They share no elements.

Teacher answer key: Primary Six Mathematics | Term 1 | Week 1 | Lesson 3: Difference and Complement of Sets

Answers to practice

- 1. {1, 2}.
- 2. {5}.
- 3. {3, 4}.
- 4. {2, 4, 5}.
- 5. {b, c}.
- 6. Inside A but outside B.
- 7. ∅.
- 8. It tells us which possible elements are being considered.

Check of worked examples

- 1. Remove 3 and 7 from A. Answer: {2, 5}.
- 2. C′ = {a, c, d}.
- 3. No. A complement contains only elements from U.

Teacher answer key: Primary Six Mathematics | Term 1 | Week 1 | Lesson 4: Subsets and the Number of Subsets

Answers to practice

- 1. Yes.
- 2. $\varnothing$, {p}.
- 3. $\varnothing$, {x}, {y}, {x, y}.
- 4. 8.
- 5. 7.
- 6. 32.
- 7. 4 because $2^4 = 16$.
- 8. 3 because $2^3 - 1 = 7$.

Check of worked examples

- 1. $\varnothing$, {1}, {2}, {1, 2}.
- 2. $2^4 = 16$ subsets.
- 3. $2^5 - 1 = 32 - 1 = 31$.

Teacher answer key: Primary Six Mathematics | Term 1 | Week 1 | Lesson 5: Finding the Number of Elements in Sets

Answers to practice

- 1. 5.
- 2. 3.
- 3. 0.
- 4. 12.
- 5. 11.
- 6. 33 (22 + 19 − 8).
- 7. 7 (40 − 33).
- 8. 16.

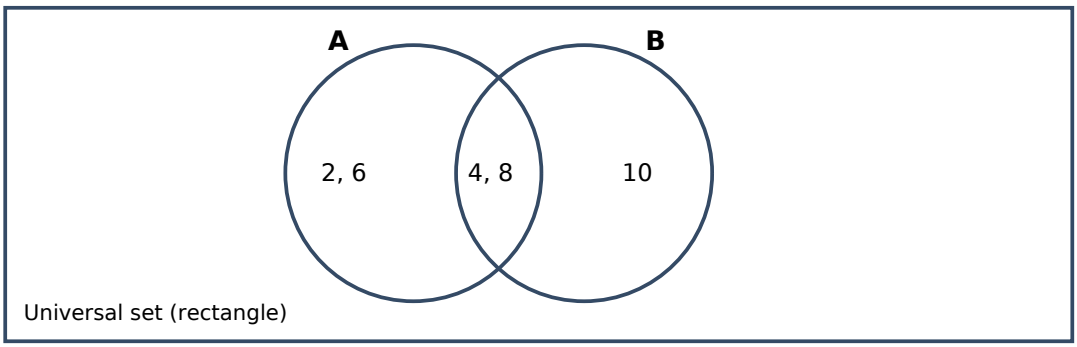
Check of worked examples

- 1. There are four distinct elements: 1, 2, 3, 4. So n(A) = 4.
- 2. 12 + 10 − 4 = 18.
- 3. At least one = 18 + 13 − 6 = 25. Neither = 30 − 25 = 5.

Primary Six Mathematics — Representing and interpreting Venn diagrams

Teacher answer key

Worked examples



1. A={2,4,6,8}, B={4,8,10}. Fill the diagram.

• Both: {4,8}; A only: {2,6}; B only: {10}.

2. Find n(A∪B).

• Count each member once: {2,4,6,8,10}, so n(A∪B)=5.

Practice answers

1. A={1,2,3}, B={2,3,4}. List the overlap. — {2,3}

2. List A only. — {1}

3. List B only. — {4}

4. Find n(A∪B). — 4

5. A={a,b,c}, B={c,d}. List the union. — {a,b,c,d}

6. Find n(A∩B). — 1

7. Draw two circles for sets of odd and prime numbers from 1 to 10. — Overlap {3,5,7}; odd only {1,9}; prime only {2}

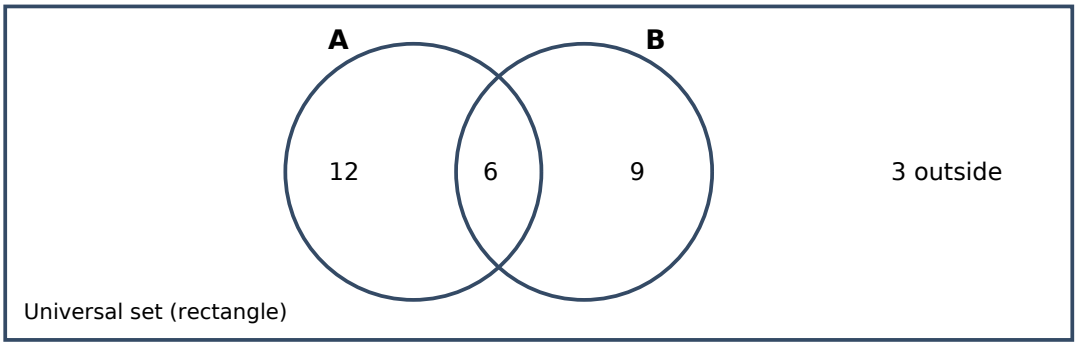
8. Why is 2 in the prime-only region? — 2 is prime and even.

Marking: Accept correct equivalent methods and simplified answers.

Primary Six Mathematics — Word problems with two sets

Teacher answer key

Worked examples



1. In 30 pupils, 18 play netball, 15 play football and 6 play both. How many play at least one?

• 18+15−6=27. Neither=30−27=3.

2. If 24 like tea, 20 like milk and 10 like both, how many like only tea?

• 24−10=14.

Practice answers

1. In a class, 14 like rice, 12 like matooke and 5 like both. Find the union. — 21

2. How many like rice only? — 9

3. How many like matooke only? — 7

4. In a group of 30, 21 study Art, 18 study Music, 12 study both. Find neither. — 3

5. If n(A)=13, n(B)=16 and n(A∩B)=4, find n(A∪B). — 25

6. If union=28, n(A)=20, n(B)=17, find intersection. — 9

7. Write the number in exactly one group if A only=7 and B only=11. — 18

8. Why must you subtract the intersection when adding group totals? — Shared members were counted twice.

Marking: Accept correct equivalent methods and simplified answers.

Primary Six Mathematics — Simple probability

Teacher answer key

Worked examples

1. A fair die is thrown. Find P(even).

• Even faces: 2,4,6. P=3/6=1/2.

2. A bag contains 3 red and 5 blue beads. Find P(red).

• 3 out of 8, so P=3/8.

Practice answers

1. P(head) for a fair coin. — 1/2

2. P(5) on a fair six-sided die. — 1/6

3. P(number greater than 4) on the die. — 1/3

4. A bag has 4 green and 6 yellow beads. P(green)? — 2/5

5. Find P(yellow). — 3/5

6. P(7) on an ordinary die? — 0

7. P(a number from 1 to 6) on an ordinary die? — 1

8. If P(rain)=2/5, find P(no rain). — 3/5

Marking: Accept correct equivalent methods and simplified answers.

Primary Six Mathematics — Place value and digit value

Teacher answer key

Worked examples

1. In 4,725,306 find the place and value of 2.

• The 2 is in the ten-thousands place; its value is 20,000.

2. Find the value of 7 in 7,092,145.

• 7,000,000.

Practice answers

1. Value of 6 in 963,102. — 60,000

2. Place of 8 in 4,817,293. — hundred-thousands

3. Value of 3 in 503,214. — 3,000

4. What digit is in the millions place in 8,657,300? — 8

5. Value of 5 in 1,530,041. — 500,000

6. Write 40,000+2,000+300+8 as a numeral. — 42,308

7. Which is greater, 506,070 or 560,007? — 560,007

8. Find the difference between digit value and place value. — Value is amount; place value is position.

Marking: Accept correct equivalent methods and simplified answers.

Primary Six Mathematics — Expanded whole numbers and powers of ten

Teacher answer key

Worked examples

1. Expand 52,304.

• 50,000+2,000+300+4.

2. Write 3,406 in powers of ten.

• $3\times10^3+4\times10^2+0\times10^1+6\times10^0$.

Practice answers

1. Expand 6,205. — 6,000+200+5

2. Expand 91,047. — 90,000+1,000+40+7

3. Find 10^4. — 10,000

4. Find 4×10^3. — 4,000

5. Write 7,030 as a sum. — 7,000+30

6. Write $8\times10^2+5\times10+2$. — 852

7. What is 10^0? — 1

8. Find $2\times10^4+3\times10^2+7$. — 20,307

Marking: Accept correct equivalent methods and simplified answers.

Primary Six Mathematics — Standard form or scientific notation

Teacher answer key

Worked examples

1. Write 42,000 in standard form.

• 4.2×10^{4}.

2. Write 0.0035 in standard form.

• 3.5×10^{-3}.

Practice answers

1. 8,000 in standard form. — 8×10^{3}

2. 63,000 in standard form. — 6.3×10^{4}

3. 0.004 in standard form. — 4×10^{-3}

4. 0.00072 in standard form. — 7.2×10^{-4}

5. 3.6×10^{3} as a numeral. — 3,600

6. 5×10^{-2} as decimal. — 0.05

7. 1,250,000 in standard form. — 1.25×10^{6}

8. Explain the range of a in standard form. — $1\leq a<10$

Marking: Accept correct equivalent methods and simplified answers.

Primary Six Mathematics — Writing expanded numbers as numerals

Teacher answer key

Worked examples

1. Find $5\times10^4+2\times10^3+6\times10+8$.

• 50,000+2,000+60+8=52,068.

2. Find 300,000+40,000+50+9.

• 340,059.

Practice answers

1. 6,000+400+30+2. — 6,432

2. $9\times10^3+3\times10^2+5$. — 9,305

3. 70,000+800+6. — 70,806

4. $2\times10^5+4\times10^3+3\times10$. — 204,030

5. $8\times10^2+2\times10+1$. — 821

6. $5\times10^4+7$. — 50,007

7. 900,000+9,000+90+9. — 909,099

8. $6\times10^3+0\times10^2+4\times10+2$. — 6,042

Marking: Accept correct equivalent methods and simplified answers.

Primary Six Mathematics — Reading and writing whole numbers in words

Teacher answer key

Worked examples

1. Write 2,304,015 in words.

• Two million, three hundred and four thousand and fifteen.

2. Write 70,021 in words.

• Seventy thousand and twenty-one.

Practice answers

1. Write 8,001 in words. — Eight thousand and one

2. Write 20,020 in words. — Twenty thousand and twenty

3. Write 304,005 in words. — Three hundred and four thousand and five

4. Write 1,000,000 in words. — One million

5. Write 42,019 in words. — Forty-two thousand and nineteen

6. Write 900,300 in words. — Nine hundred thousand, three hundred

7. Write 14,070 in words. — Fourteen thousand and seventy

8. Write 5,050,500 in words. — Five million, fifty thousand, five hundred

Marking: Accept correct equivalent methods and simplified answers.

Primary Six Mathematics — Changing number words to figures

Teacher answer key

Worked examples

1. Write “three million, forty thousand and six” in figures.

• 3,040,006.

2. Write “six hundred and two thousand and seventy” in figures.

• 602,070.

Practice answers

1. Eight thousand and nine. — 8,009

2. Seventy thousand and fifty. — 70,050

3. Three hundred and ten thousand. — 310,000

4. Two million, six thousand and four. — 2,006,004

5. Nine hundred thousand and five. — 900,005

6. Twenty-five thousand, one hundred and four. — 25,104

7. Four million and thirty. — 4,000,030

8. Five hundred and six thousand, seven hundred. — 506,700

Marking: Accept correct equivalent methods and simplified answers.

Primary Six Mathematics — Rounding whole numbers

Teacher answer key

Worked examples

1. Round 47,682 to the nearest thousand.

• Hundreds digit is 6, so 48,000.

2. Round 635,124 to nearest ten thousand.

• Thousands digit is 5, so 640,000.

Practice answers

1. 368 to nearest ten. — 370

2. 4,249 to nearest hundred. — 4,200

3. 7,950 to nearest thousand. — 8,000

4. 82,499 to nearest ten thousand. — 80,000

5. 75 to nearest ten. — 80

6. 14,501 to nearest thousand. — 15,000

7. 999 to nearest hundred. — 1,000

8. 6,544 to nearest ten. — 6,540

Marking: Accept correct equivalent methods and simplified answers.

Primary Six Mathematics — Decimal place values

Teacher answer key

Worked examples

1. Find the value of 8 in 6.084.

• 8 hundredths = 0.08.

2. Write 3+2/10+5/100 as decimal.

• 3.25.

Practice answers

1. Value of 4 in 12.405. — 0.4

2. Value of 7 in 0.078. — 0.07

3. Place of 9 in 5.009. — thousandths

4. 3+6/10+2/100. — 3.62

5. Write 0.5 as a fraction. — 1/2

6. Write 0.04 as a fraction. — 1/25

7. Find 4 ones+6 hundredths. — 4.06

8. Compare 0.7 and 0.65. — 0.7 is greater

Marking: Accept correct equivalent methods and simplified answers.

Primary Six Mathematics — Writing decimals in words and figures

Teacher answer key

Worked examples

1. Write 6.25 in words.

• Six and twenty-five hundredths.

2. Write “three and seven thousandths” as a decimal.

• 3.007.

Practice answers

1. Write 0.6 in words. — Six tenths

2. Write 2.04 in words. — Two and four hundredths

3. Nine and five tenths in figures. — 9.5

4. Four and eight hundredths in figures. — 4.08

5. Seven thousandths in figures. — 0.007

6. Write 10.01 in words. — Ten and one hundredth

7. One and twelve hundredths in figures. — 1.12

8. Write 0.205 in words. — Two hundred and five thousandths

Marking: Accept correct equivalent methods and simplified answers.

Primary Six Mathematics — Expanded form of decimals

Teacher answer key

Worked examples

1. Expand 7.438.

• 7+0.4+0.03+0.008.

2. Expand 52.06.

• 50+2+0.06.

Practice answers

1. Expand 3.25. — 3+0.2+0.05

2. Expand 0.407. — 0.4+0.007

3. Expand 15.08. — 10+5+0.08

4. Expand 2.006. — 2+0.006

5. Express 4.32 using fractions. — 4+3/10+2/100

6. Expand 0.91. — 0.9+0.01

7. Expand 8.005. — 8+0.005

8. Expand 70.304. — 70+0.3+0.004

Marking: Accept correct equivalent methods and simplified answers.

Primary Six Mathematics — Finding decimals from expanded form

Teacher answer key

Worked examples

1. Find 5+0.2+0.007.

• 5.207.

2. Find 30+4+0.06+0.003.

• 34.063.

Practice answers

1. 2+0.4+0.05. — 2.45

2. 8+0.003. — 8.003

3. 0.6+0.08. — 0.68

4. 10+0.9+0.004. — 10.904

5. 7+5/100. — 7.05

6. 1+4/10+6/1000. — 1.406

7. 20+0.01+0.008. — 20.018

8. 3+0.7+0.02+0.005. — 3.725

Marking: Accept correct equivalent methods and simplified answers.

Primary Six Mathematics — Ordering decimals

Teacher answer key

Worked examples

1. Order 0.9, 0.85 and 0.805 from smallest.

• 0.805, 0.85, 0.9 (write 0.900 for comparison).

2. Which is greater, 3.07 or 3.7?

• 3.7 because 3.70>3.07.

Practice answers

1. Ascending: 0.2, 0.12, 0.21. — 0.12,0.2,0.21

2. Descending: 4.04,4.4,4.004. — 4.4,4.04,4.004

3. Compare 0.80 and 0.8. — Equal

4. Compare 1.09 and 1.1. — 1.1 greater

5. Smallest: 0.06,0.6,0.606. — 0.06

6. Largest: 2.003,2.03,2.3. — 2.3

7. Ascending: 5.1,5.01,5.001. — 5.001,5.01,5.1

8. Insert < or >: 0.507 __ 0.57. — <

Marking: Accept correct equivalent methods and simplified answers.

Primary Six Mathematics — Rounding decimals

Teacher answer key

Worked examples

1. Round 5.678 to 2 decimal places.

• Third decimal is 8: 5.68.

2. Round 9.46 to 1 decimal place.

• Hundredths digit is 6: 9.5.

Practice answers

1. 3.74 to 1 dp. — 3.7

2. 3.75 to 1 dp. — 3.8

3. 0.986 to 2 dp. — 0.99

4. 4.005 to 2 dp. — 4.01

5. 8.499 to whole number. — 8

6. 8.5 to whole number. — 9

7. 12.347 to 2 dp. — 12.35

8. 0.444 to 2 dp. — 0.44

Marking: Accept correct equivalent methods and simplified answers.

Primary Six Mathematics — Roman numerals

Teacher answer key

Worked examples

1. Write 49 in Roman numerals.

• XLIX = 40+9.

2. Convert CDXLVI to figures.

• 400+40+6=446.

Practice answers

1. 19 in Roman numerals. — XIX

2. 40 in Roman numerals. — XL

3. 90 in Roman numerals. — XC

4. 99 in Roman numerals. — XCIX

5. 144 in Roman numerals. — CXLIV

6. MM in figures. — 2000

7. DCC in figures. — 700

8. CMIV in figures. — 904

Marking: Accept correct equivalent methods and simplified answers.

Primary Six Mathematics — Operations with Roman numerals

Teacher answer key

Worked examples

1. Calculate XII+IX.

• 12+9=21=XXI.

2. Calculate XL−XIV.

• 40−14=26=XXVI.

Practice answers

1. V+VI. — XI

2. XX−VIII. — XII

3. X+XV. — XXV

4. L−X. — XL

5. IX+IX. — XVIII

6. C−L. — L

7. XL+X. — L

8. XXX−XIX. — XI

Marking: Accept correct equivalent methods and simplified answers.

Primary Six Mathematics — Counting in base two and base five

Teacher answer key

Worked examples

1. Change 1011_2 to base ten.

• $1\times8+0\times4+1\times2+1=11$.

2. Change 23_5 to base ten.

• $2\times5+3=13$.

Practice answers

1. 10_2 in decimal. — 2

2. 111_2 in decimal. — 7

3. 100_2 in decimal. — 4

4. 14_5 in decimal. — 9

5. 32_5 in decimal. — 17

6. 20_5 in decimal. — 10

7. Write 5 in base two. — 101_2

8. Write 8 in base five. — 13_5

Marking: Accept correct equivalent methods and simplified answers.

Primary Six Mathematics — Adding numbers in other bases

Teacher answer key

Worked examples

1. Add 23_5+14_5.

• $3+4=7=12_5$: put 2, carry 1; $2+1+1=4$; answer 42_5.

2. Add 101_2+11_2.

• $5+3=8=1000_2$.

Practice answers

1. 12_5+11_5. — 23_5

2. 24_5+1_5. — 30_5

3. 101_2+1_2. — 110_2

4. 11_2+11_2. — 110_2

5. 13_5+12_5. — 30_5

6. 110_2+10_2. — 1000_2

7. 21_5+3_5. — 24_5

8. 100_2+101_2. — 1001_2

Marking: Accept correct equivalent methods and simplified answers.

Primary Six Mathematics — Subtracting and multiplying in other bases

Teacher answer key

Worked examples

1. Subtract $43_5 - 14_5$.

• $23 - 9 = 14$ decimal, which is 24_5.

2. Multiply $12_5 \times 3_5$.

• $7 \times 3 = 21$ decimal $= 41_5$.

Practice answers

1. $24_5 - 3_5$. — 21_5

2. $30_5 - 1_5$. — 24_5

3. $110_2 - 10_2$. — 100_2

4. $101_2 - 11_2$. — 10_2

5. $4_5 \times 2_5$. — 13_5

6. $11_2 \times 10_2$. — 110_2

7. $12_5 \times 2_5$. — 24_5

8. $100_2 \times 11_2$. — 1100_2

Marking: Accept correct equivalent methods and simplified answers.

Primary Six Mathematics — Finite number systems and clock arithmetic

Teacher answer key

Worked examples

1. What is 9+7 on a 12-hour clock?

• 16; subtract 12, giving 4.

2. A 7-day cycle begins Monday. What day is 10 days later?

• 10 leaves remainder 3 after 7; Thursday.

Practice answers

1. On 12-hour clock, 8+6. — 2

2. On 12-hour clock, 11+3. — 2

3. On 12-hour clock, 2+10. — 12

4. On 7-day cycle, two days after Friday. — Sunday

5. On 7-day cycle, seven days after Tuesday. — Tuesday

6. On 5-step cycle labelled 0–4, 3+4. — 2

7. On 5-step cycle labelled 0–4, 4+4. — 3

8. Why do remainders help? — They show the position after complete cycles.

Marking: Accept correct equivalent methods and simplified answers.

Primary Six Mathematics — Subtracting with finite cycles

Teacher answer key

Worked examples

1. On a 12-hour clock, calculate 2−7.

• Go back 7 hours from 2: 7 o’clock.

2. In a 7-day cycle, what day was four days before Wednesday?

• Saturday.

Practice answers

1. 12-hour clock: 4−6. — 10

2. 12-hour clock: 9−11. — 10

3. 12-hour clock: 5−3. — 2

4. 12-hour clock: 1−2. — 11

5. 3 days before Monday. — Friday

6. 5 days before Thursday. — Saturday

7. On 5-step cycle 0–4: 1−3. — 3

8. On 5-step cycle 0–4: 2−4. — 3

Marking: Accept correct equivalent methods and simplified answers.

Primary Six Mathematics — Unknowns in finite systems

Teacher answer key

Worked examples

1. Find x between 1 and 12 if x+5 gives 3 on a 12-hour clock.

• x+5=15, so x=10.

2. Find x in {0,1,2,3,4} if x+4 gives 2 in a 5-step cycle.

• x=3 because 3+4=7, remainder 2.

Practice answers

1. 12-hour cycle: x+3=2. — 11

2. 12-hour cycle: x+7=4. — 9

3. 12-hour cycle: x−5=2. — 7

4. 5-step cycle: x+2=0. — 3

5. 5-step cycle: x+1=4. — 3

6. 7-step cycle 0–6: x+3=1. — 5

7. 12-hour cycle: x+9=12. — 3

8. 5-step cycle: x−2=4. — 1

Marking: Accept correct equivalent methods and simplified answers.

Primary Six Mathematics — Applications of finite cycles

Teacher answer key

Worked examples

1. Today is Thursday. What day will it be in 17 days?

• 17=14+3: Sunday.

2. A clock shows 9 o’clock. What time will it show 28 hours later?

• 28=24+4: 1 o’clock.

Practice answers

1. 10 days after Monday. — Thursday

2. 15 days after Friday. — Saturday

3. 24 hours after 6 o’clock. — 6 o’clock

4. 14 days before Sunday. — Sunday

5. 49 days after Wednesday. — Wednesday

6. 20 hours after 8 o’clock. — 4 o’clock

7. 9 days before Tuesday. — Sunday

8. 30 hours after 5 o’clock. — 11 o’clock

Marking: Accept correct equivalent methods and simplified answers.

Primary Six Mathematics — Adding whole numbers

Teacher answer key

Worked examples

1. Add 38,745+26,589.

• 38,745+26,589=65,334.

2. A trader sells 2,475 oranges on Monday and 1,988 on Tuesday. How many?

• 2,475+1,988=4,463 oranges.

Practice answers

1. 468+257. — 725

2. 2,345+1,876. — 4,221

3. 87,600+9,405. — 97,005

4. 625+375. — 1,000

5. 39,900+101. — 40,001

6. 14,325+2,500. — 16,825

7. 12,345+54,321. — 66,666

8. 3,067+2,908. — 5,975

Marking: Accept correct equivalent methods and simplified answers.

Primary Six Mathematics — Subtracting whole numbers

Teacher answer key

Worked examples

1. Subtract 28,476−9,858.

• 18,618.

2. A school has 1,250 exercise books and gives out 785. How many remain?

• 1,250−785=465 books.

Practice answers

1. 700−258. — 442

2. 5,000−1,999. — 3,001

3. 43,105−2,678. — 40,427

4. 1,204−589. — 615

5. 90,000−800. — 89,200

6. 726−447. — 279

7. 35,100−12,905. — 22,195

8. 6,302−3,408. — 2,894

Marking: Accept correct equivalent methods and simplified answers.

Primary Six Mathematics — Multiplication of whole numbers

Teacher answer key

Worked examples

1. Multiply 325×56.

• 325×(50+6)=16,250+1,950=18,200.

2. A box holds 24 pencils. How many in 36 boxes?

• 24×36=864 pencils.

Practice answers

1. 45×23. — 1,035

2. 126×14. — 1,764

3. 208×35. — 7,280

4. 125×8. — 1,000

5. 304×12. — 3,648

6. 67×19. — 1,273

7. 250×24. — 6,000

8. 111×111. — 12,321

Marking: Accept correct equivalent methods and simplified answers.

Primary Six Mathematics — Long division of whole numbers

Teacher answer key

Worked examples

1. Divide 3,684 by 12.

• 12×307=3,684, so quotient 307.

2. Share 1,248 mangoes equally among 24 shops.

• 1,248÷24=52 mangoes per shop.

Practice answers

1. 936÷12. — 78

2. 1,008÷24. — 42

3. 1,440÷36. — 40

4. 805÷5. — 161

5. 2,304÷48. — 48

6. 4,200÷70. — 60

7. 2,475÷25. — 99

8. 6,048÷12. — 504

Marking: Accept correct equivalent methods and simplified answers.

Primary Six Mathematics — Division word problems

Teacher answer key

Worked examples

1. A school shares 1,176 books equally among 28 classes. Find books per class.

• 1,176÷28=42 books.

2. A farmer packs 975 eggs in trays of 25. How many trays?

• 975÷25=39 trays.

Practice answers

1. 540 learners travel equally in 12 buses. Learners per bus? — 45

2. 756 pencils shared among 21 pupils. Each gets? — 36

3. 840 kg divided into 35 sacks. kg per sack? — 24

4. 1,000 shillings shared equally by 8 children. Each gets? — 125 UGX

5. 720 books packed 24 per carton. Number of cartons? — 30

6. 1,260 litres into 42 drums. Each drum? — 30 litres

7. How many groups of 15 in 435? — 29

8. A farmer has 924 seedlings in 22 rows. Seedlings per row? — 42

Marking: Accept correct equivalent methods and simplified answers.

Primary Six Mathematics — Mixed operations and BODMAS

Teacher answer key

Worked examples

1. Calculate $8+6\times3$.

• Multiply first: $8+18=26$.

2. Calculate $(18-6)\div3+2^2$.

• $12\div3+4=8$.

Practice answers

1. $4+5\times6$. — 34

2. $(4+5)\times6$. — 54

3. $24\div6\times2$. — 8

4. $36-8\div2$. — 32

5. 3^2+7. — 16

6. $20-(6+4)$. — 10

7. $5\times(3+2)-4$. — 21

8. $12+18\div3-2$. — 16

Marking: Accept correct equivalent methods and simplified answers.

Primary Six Mathematics — Number operation properties

Teacher answer key

Worked examples

1. Use distributive property for 7×(20+3).

• 7×20+7×3=140+21=161.

2. Use associative property for (4×25)×3.

• 4×(25×3)=300.

Practice answers

1. 8+5=5+___. — 8

2. 6×9=9×___. — 6

3. 3×(10+4). — 42

4. (2+3)+7=2+(3+___). — 7

5. 5×(8+2). — 50

6. 4×(20−3). — 68

7. Can subtraction be commutative? — No

8. Name property: 2×(3+4)=2×3+2×4. — Distributive

Marking: Accept correct equivalent methods and simplified answers.

Primary Six Mathematics — Divisibility tests

Teacher answer key

Worked examples

1. Is 4,356 divisible by 9?

• 4+3+5+6=18, which is divisible by 9, so yes.

2. Is 2,148 divisible by 4?

• Last two digits 48 divisible by 4, so yes.

Practice answers

1. Is 345 divisible by 3? — Yes

2. Is 345 divisible by 9? — No

3. Is 2,160 divisible by 10? — Yes

4. Is 2,160 divisible by 8? — Yes

5. Is 234 divisible by 6? — Yes

6. Is 1,025 divisible by 5? — Yes

7. Is 148 divisible by 4? — Yes

8. Is 237 divisible by 2? — No

Marking: Accept correct equivalent methods and simplified answers.

Primary Six Mathematics — Developing number patterns

Teacher answer key

Worked examples

1. Continue 3,6,10,15, __.

• Differences 3,4,5, so next difference 6: 21.

2. Continue 2,4,8,16, __.

• Multiply by 2: 32.

Practice answers

1. 2,5,8,11,___. — 14

2. 1,4,9,16,___. — 25

3. 1,3,6,10,___. — 15

4. 40,35,30,___. — 25

5. 3,9,27,___. — 81

6. 4,8,12,16,___. — 20

7. 1,2,4,7,11,___. — 16

8. 100,50,25,___. — 12.5

Marking: Accept correct equivalent methods and simplified answers.

Primary Six Mathematics — Prime and composite numbers

Teacher answer key

Worked examples

1. Is 29 prime?

• Yes. It has factors 1 and 29 only.

2. Is 21 composite?

• Yes. 21=3×7, so it has more than two factors.

Practice answers

1. Is 2 prime? — Yes

2. Is 1 prime? — No

3. Is 9 composite? — Yes

4. List primes below 10. — 2,3,5,7

5. Is 31 prime? — Yes

6. Is 49 composite? — Yes

7. Smallest composite number? — 4

8. Is 37 prime? — Yes

Marking: Accept correct equivalent methods and simplified answers.

Primary Six Mathematics — Consecutive whole numbers

Teacher answer key

Worked examples

1. Three consecutive whole numbers sum to 48. Find them.

• Let them be n,n+1,n+2. 3n+3=48; n=15. Numbers 15,16,17.

2. Find two consecutive numbers whose sum is 29.

• n+(n+1)=29; n=14, so 14 and 15.

Practice answers

1. Numbers after 19 and 20. — 21,22

2. Three consecutive numbers beginning at 7. — 7,8,9

3. Sum of 8,9,10. — 27

4. Two consecutive numbers sum to 51. — 25,26

5. Three consecutive numbers sum to 36. — 11,12,13

6. Write three consecutive numbers after n. — n+1,n+2,n+3

7. Find n if n+(n+1)=17. — 8

8. Are 5,7,9 consecutive whole numbers? — No

Marking: Accept correct equivalent methods and simplified answers.

Primary Six Mathematics — Consecutive odd and even numbers

Teacher answer key

Worked examples

1. Three consecutive even numbers sum to 42.

• n+(n+2)+(n+4)=42; n=12. Numbers 12,14,16.

2. Two consecutive odd numbers sum to 40.

• n+(n+2)=40; n=19, giving 19,21.

Practice answers

1. Next two evens after 18. — 20,22

2. Next two odds after 21. — 23,25

3. Sum of 5,7,9. — 21

4. Two consecutive evens sum to 26. — 12,14

5. Two consecutive odds sum to 32. — 15,17

6. Three consecutive even numbers sum to 60. — 18,20,22

7. Three consecutive odd numbers sum to 45. — 13,15,17

8. Difference between consecutive odd numbers. — 2

Marking: Accept correct equivalent methods and simplified answers.

Primary Six Mathematics — Factors and highest common factor

Teacher answer key

Worked examples

1. Find HCF of 24 and 36.

• Factors of 24: 1,2,3,4,6,8,12,24. Factors of 36 include 1,2,3,4,6,9,12,18,36. HCF=12.

2. Find HCF of 18 and 30.

• Common factors 1,2,3,6; HCF=6.

Practice answers

1. Factors of 10. — 1,2,5,10

2. HCF of 12 and 18. — 6

3. HCF of 15 and 25. — 5

4. HCF of 7 and 9. — 1

5. HCF of 16 and 24. — 8

6. HCF of 21 and 28. — 7

7. Is 6 a factor of 42? — Yes

8. Largest factor of 29? — 29

Marking: Accept correct equivalent methods and simplified answers.

Primary Six Mathematics — Prime factorisation

Teacher answer key

Worked examples

1. Prime-factorise 72.

• $72=8\times9=2\times2\times2\times3\times3=2^3\times3^2$.

2. Prime-factorise 90.

• $90=2\times3\times3\times5=2\times3^2\times5$.

Practice answers

1. Prime factors of 12. — $2^2\times3$

2. Prime factors of 20. — $2^2\times5$

3. Prime factors of 36. — $2^2\times3^2$

4. Prime factors of 50. — 2×5^2

5. Prime factors of 48. — $2^4\times3$

6. Prime factors of 100. — $2^2\times5^2$

7. Prime factors of 27. — 3^3

8. Prime factors of 84. — $2^2\times3\times7$

Marking: Accept correct equivalent methods and simplified answers.

Primary Six Mathematics — Finding numbers from prime factors

Teacher answer key

Worked examples

1. Find $2^3 \times 3^2$.

• $8 \times 9 = 72$.

2. Find $2^2 \times 5^2$.

• $4 \times 25 = 100$.

Practice answers

1. 2×3^2. — 18

2. $2^3 \times 5$. — 40

3. $3^2 \times 7$. — 63

4. $2^4 \times 3$. — 48

5. $5^2 \times 2$. — 50

6. $2^2 \times 3 \times 5$. — 60

7. $3^3 \times 2$. — 54

8. 2×7^2. — 98

Marking: Accept correct equivalent methods and simplified answers.

Primary Six Mathematics — Multiples and lowest common multiple

Teacher answer key

Worked examples

1. Find LCM of 6 and 8.

• Multiples of 6: 6,12,18,24; of 8: 8,16,24. LCM=24.

2. Find LCM of 9 and 12.

• 9:9,18,27,36; 12:12,24,36. LCM=36.

Practice answers

1. LCM of 4 and 6. — 12

2. LCM of 5 and 10. — 10

3. LCM of 6 and 9. — 18

4. LCM of 3 and 7. — 21

5. LCM of 8 and 12. — 24

6. LCM of 7 and 14. — 14

7. LCM of 10 and 15. — 30

8. List first three multiples of 9. — 9,18,27

Marking: Accept correct equivalent methods and simplified answers.

Primary Six Mathematics — HCF and LCM using prime factors

Teacher answer key

Worked examples

1. Find HCF and LCM of 12 and 18.

• $12=2^2\times3$; $18=2\times3^2$. HCF$=2\times3=6$; LCM$=2^2\times3^2=36$.

2. Find HCF and LCM of 20 and 30.

• $20=2^2\times5$; $30=2\times3\times5$. HCF$=10$; LCM$=60$.

Practice answers

1. HCF,LCM of 8 and 12. — 4,24

2. HCF,LCM of 6 and 15. — 3,30

3. HCF,LCM of 9 and 12. — 3,36

4. HCF,LCM of 10 and 25. — 5,50

5. HCF,LCM of 14 and 21. — 7,42

6. HCF,LCM of 16 and 20. — 4,80

7. HCF,LCM of 18 and 24. — 6,72

8. Check HCF×LCM for 12 and 18. — 216=12×18

Marking: Accept correct equivalent methods and simplified answers.

Primary Six Mathematics — Finding unknown prime factors

Teacher answer key

Worked examples

1. Two numbers are 12 and x. Their HCF is 6 and LCM 60. Find x.

• $6\times60=12\times x$; $x=30$.

2. If $36=2^2\times3^2$, which prime factor is missing in $2\times2\times3\times__$?

• 3.

Practice answers

1. Find x if HCF=3, LCM=24, a=6. — 12

2. Find x if HCF=5, LCM=60, a=15. — 20

3. Missing factor: $30=2\times3\times__$. — 5

4. Missing factor: $84=2\times2\times3\times__$. — 7

5. HCF of 18 and 27. — 9

6. LCM of 18 and 27. — 54

7. HCF×LCM for 8 and 12. — 96

8. Find b if HCF=4, LCM=40, a=8. — 20

Marking: Accept correct equivalent methods and simplified answers.

Primary Six Mathematics — Applying HCF and LCM

Teacher answer key

Worked examples

1. Ropes of 24 m and 36 m must be cut into longest equal pieces. Length?

• HCF(24,36)=12 m.

2. Two bells ring every 6 and 8 minutes. When together next?

• LCM(6,8)=24 minutes.

Practice answers

1. Longest equal pieces from 18 m and 30 m. — 6 m

2. Bells every 4 and 10 min: next together. — 20 min

3. HCF of 16 and 40. — 8

4. LCM of 9 and 15. — 45

5. Maximum equal groups from 24 and 32 items. — 8 groups

6. Bus visits every 12 and 18 days: next same day. — 36 days

7. HCF of 28 and 42. — 14

8. LCM of 5 and 6. — 30

Marking: Accept correct equivalent methods and simplified answers.

Primary Six Mathematics — Word problems with common multiples

Teacher answer key

Worked examples

1. A bus comes every 15 min and a van every 20 min. Both come at 8:00. Next together?

• LCM(15,20)=60 min, so 9:00.

2. Lights flash every 8 s and 12 s. When together again?

• LCM(8,12)=24 s.

Practice answers

1. Bells every 6 and 9 min. — 18 min

2. Buses every 10 and 25 min. — 50 min

3. Lights every 4 and 7 s. — 28 s

4. Two cycles 12 and 15 days. — 60 days

5. Bells every 5 and 8 min. — 40 min

6. Two clocks every 20 and 30 min. — 60 min

7. Lights every 6 and 10 seconds. — 30 seconds

8. Intervals 9 and 12 min starting 10:00. Next same time? — 10:36

Marking: Accept correct equivalent methods and simplified answers.

Primary Six Mathematics — Powers of whole numbers

Teacher answer key

Worked examples

1. Simplify $2^3 \times 2^4$.

• $2^7 = 128$.

2. Simplify $5^4 \div 5^2$.

• $5^2 = 25$.

Practice answers

1. 3^2. — 9

2. 2^5. — 32

3. 4^3. — 64

4. 10^3. — 1000

5. $2^2 \times 2^3$. — 32

6. $3^4 \div 3^2$. — 9

7. 5^0. — 1

8. $(2^2)^3$. — 64

Marking: Accept correct equivalent methods and simplified answers.

Primary Six Mathematics — Squares of whole numbers and fractions

Teacher answer key

Worked examples

1. Find $(3/4)^2$.

• 9/16.

2. Find 1.2^2.

• $1.2 \times 1.2 = 1.44$.

Practice answers

1. 6^2. — 36

2. 11^2. — 121

3. $(2/3)^2$. — 4/9

4. $(1/5)^2$. — 1/25

5. 0.5^2. — 0.25

6. 1.5^2. — 2.25

7. 15^2. — 225

8. $(1\ 1/2)^2$. — 2 1/4

Marking: Accept correct equivalent methods and simplified answers.

Primary Six Mathematics — Square roots of whole numbers

Teacher answer key

Worked examples

1. Find √144.

• 12×12=144, so √144=12.

2. Find √225.

• 15×15=225, so √225=15.

Practice answers

1. √25. — 5

2. √64. — 8

3. √81. — 9

4. √121. — 11

5. √169. — 13

6. √196. — 14

7. √400. — 20

8. √1. — 1

Marking: Accept correct equivalent methods and simplified answers.

Primary Six Mathematics — Square roots of fractions and decimals

Teacher answer key

Worked examples

1. Find √(9/16).

• √9/√16=3/4.

2. Find √0.49.

• 0.7×0.7=0.49; answer 0.7.

Practice answers

1. √(1/4). — 1/2

2. √(25/36). — 5/6

3. √0.09. — 0.3

4. √0.64. — 0.8

5. √(49/100). — 7/10

6. √2.25. — 1.5

7. √(4/9). — 2/3

8. √0.01. — 0.1

Marking: Accept correct equivalent methods and simplified answers.

Primary Six Mathematics — Applying square roots

Teacher answer key

Worked examples

1. A square field has area 196 m^2. Find its side.

• $\sqrt{196}=14$ m.

2. A square tile has area 0.81 m^2. Find the side.

• $\sqrt{0.81}=0.9$ m.

Practice answers

1. Square area 64 cm^2: side. — 8 cm

2. Square area 225 m^2: side. — 15 m

3. Square area 121 cm^2: side. — 11 cm

4. Square area 0.25 m^2: side. — 0.5 m

5. Square side 7 cm: area. — 49 cm^2

6. Square perimeter 40 m: side. — 10 m

7. Square area 400 cm^2: perimeter. — 80 cm

8. Side 12 m: area. — 144 m^2

Marking: Accept correct equivalent methods and simplified answers.

Primary Six Mathematics — Cubes and cube roots

Teacher answer key

Worked examples

1. Find 5^3.

• $5\times5\times5=125$.

2. Find $\sqrt[3]{216}$.

• $6^3=216$, so $\sqrt[3]{216}=6$.

Practice answers

1. 2^3. — 8

2. 3^3. — 27

3. 4^3. — 64

4. 10^3. — 1000

5. $\sqrt[3]{8}$. — 2

6. $\sqrt[3]{27}$. — 3

7. $\sqrt[3]{125}$. — 5

8. $\sqrt[3]{1000}$. — 10

Marking: Accept correct equivalent methods and simplified answers.

Primary Six Mathematics — Sequences and number rules

Teacher answer key

Worked examples

1. Find the next two terms: 5,10,20,40,...

• Multiply by 2: 80,160.

2. Fill 2,6,12,20,___.

• Differences 4,6,8, so next difference 10: 30.

Practice answers

1. 7,14,21,___. — 28

2. 81,27,9,___. — 3

3. 2,5,10,17,___. — 26

4. 1,8,27,___. — 64

5. 12,17,22,___. — 27

6. 3,6,12,24,___. — 48

7. 100,90,80,___. — 70

8. 1,4,7,10,___. — 13

Marking: Accept correct equivalent methods and simplified answers.

Primary Six Mathematics — Number puzzles and magic squares

Teacher answer key

Worked examples

1. Find missing entry in a row: 8,1,__ where sum is 15.

• 15−8−1=6.

2. Why is 5 in the centre of the usual 1–9 magic square?

• The middle entry balances the rows and diagonals; the standard arrangement has 5 at the centre.

Practice answers

1. Complete a row: 4,9,__ to total 15. — 2

2. Complete a row: 3,5,__ to total 15. — 7

3. Complete a row: 2,7,__ to total 15. — 6

4. Sum of digits 1 through 9. — 45

5. Magic sum for the normal 3×3 square. — 15

6. Centre of the normal 3×3 square. — 5

7. Check 8+1+6. — 15

8. What must be true of every main diagonal? — It must sum to 15

Marking: Accept correct equivalent methods and simplified answers.